High-order Gauß time stepping for the condensed HDG beam-network problem by diagonalizing the Butcher matrix (Tricks 1 & 2)

This note specializes the decoupling of `trick1.tex` to the hybridizable DG discretization of the Timoshenko beam network of `HDG_Wave/main.tex`. It replaces the earlier generic model $\dot{\mathbf{u}} = \mathbf{C}\mathbf{u}+\mathbf{g}$ by the actual dual-mixed, *second order in time* setting, and shows that the diagonalized Gauß step reduces, once the stage factor $\Delta t\,\theta_\ell$ is fixed, to *one condensed bilinear form on the skeleton per stage* — assembled through the same mass-shifted local solvers as the fully discrete Crank–Nicolson scheme of `main.tex`, of which it is the high-order generalization. The essential point is the *order* of operations: the spatial condensation is performed *after* the time/stage discretization, so that the edge-interior inertia is kept inside the local solve. Condensing the mass *before* the time discretization (the "second-order nodal ODE" of an earlier draft) is a static condensation of the mass; it discards the edge-interior inertial modes and caps the spatial convergence at order two. Throughout, $\tau > 0$ is the HDG stabilization parameter (as in `main.tex`), the time step is Δt, and the eigenvalues of the Butcher matrix are written θ_ℓ, since λ, μ, ν already denote the hybrid unknown, its test function, and the endpoint orientation.

1 The semidiscrete problem as a first-order field system

We recall the discrete hybrid dual-mixed problem of `main.tex`. On the network $\Sigma = \bigcup_{\mathfrak{e}\in\mathcal{E}} \mathfrak{e}$ one seeks the edgewise dual and primal fields and the nodal (hybrid) unknown

$$\boldsymbol{q} = (\boldsymbol{n}, \boldsymbol{m}), \qquad \boldsymbol{y} = (\boldsymbol{u}, \boldsymbol{r}), \qquad \boldsymbol{\lambda} \in \boldsymbol{\Lambda} = \{\, \boldsymbol{\lambda} : \mathcal{N} \to \mathbb{R}^6 \,\},$$

governed, for all discrete test functions, by

$$a(\boldsymbol{q}, \boldsymbol{w}_{\mathrm{q}}) + b(\boldsymbol{w}_{\mathrm{q}}, \boldsymbol{y}) = -\langle \boldsymbol{\lambda}, \boldsymbol{w}_{\mathrm{q}}\, \nu\rangle_{\mathcal{E}}, \tag{1a}$$

$$-\,b(\boldsymbol{q}, \boldsymbol{w}_{\mathrm{y}}) + c(\partial_{tt}\boldsymbol{y}, \boldsymbol{w}_{\mathrm{y}}) + \langle \tau(\boldsymbol{y} - \boldsymbol{\lambda}), \boldsymbol{w}_{\mathrm{y}}\rangle_{\mathcal{E}} = (\boldsymbol{f}, \boldsymbol{w}_{\mathrm{y}})_{\Sigma}, \tag{1b}$$

with the compliance form a (SPD, no derivatives), the equilibrium coupling b, and the inertia form $c(\boldsymbol{y}, \boldsymbol{w}) = (\mathbf{C}_{\mathrm{y}}\boldsymbol{y}, \boldsymbol{w})_{\Sigma}$ (SPD, $\mathbf{C}_{\mathrm{y}}$ the material inertia coefficient). Two facts drive hybridization: (1) restricted to one edge is an *element-local* (edgewise) problem, and the only time derivative is the second one, $\partial_{tt}\boldsymbol{y}$; both $\boldsymbol{q}$ and the nodal $\boldsymbol{\lambda}$ are algebraic.

First-order field reformulation. Exactly as in `main.tex`, we reduce the order in time by introducing the *velocity/momentum field*

$$\boldsymbol{z} := \mathbf{C}_{\mathrm{y}}\, \partial_t \boldsymbol{y}, \qquad d(\boldsymbol{z}, \boldsymbol{w}) := (\mathbf{C}_{\mathrm{y}}^{-1}\boldsymbol{z}, \boldsymbol{w})_{\Sigma}, \tag{2}$$

so that $(\partial_t \boldsymbol{z}, \boldsymbol{w})_\Sigma = c(\partial_{tt}\boldsymbol{y}, \boldsymbol{w})$ and $d(\boldsymbol{z}, \boldsymbol{z}) = c(\partial_t \boldsymbol{y}, \partial_t \boldsymbol{y})$ is the kinetic energy. The saddle system (1) becomes the first-order-in-time system: find $(\boldsymbol{q}, \boldsymbol{y}, \boldsymbol{z}, \boldsymbol{\lambda})$ with, for all test functions,

$$
\begin{aligned}
a(\boldsymbol{q}, \boldsymbol{w}_\mathrm{q}) + b(\boldsymbol{w}_\mathrm{q}, \boldsymbol{y}) &= -\langle \boldsymbol{\lambda}, \boldsymbol{w}_\mathrm{q}\, \nu\rangle_\mathcal{E}, && \text{(3a)}\\
-\, b(\boldsymbol{q}, \boldsymbol{w}_\mathrm{y}) + \langle \tau(\boldsymbol{y} - \boldsymbol{\lambda}), \boldsymbol{w}_\mathrm{y}\rangle_\mathcal{E} + (\partial_t \boldsymbol{z}, \boldsymbol{w}_\mathrm{y})_\Sigma &= (\boldsymbol{f}, \boldsymbol{w}_\mathrm{y})_\Sigma, && \text{(3b)}\\
-\, (\partial_t \boldsymbol{y}, \boldsymbol{w}_\mathrm{z})_\Sigma + d(\boldsymbol{z}, \boldsymbol{w}_\mathrm{z})_\Sigma &= 0, && \text{(3c)}
\end{aligned}
$$

together with the nodal flux-balance condition that determines $\boldsymbol{\lambda}$. Now *two* edge fields carry time derivatives, $\boldsymbol{y}$ and $\boldsymbol{z}$; $\boldsymbol{q}$ and $\boldsymbol{\lambda}$ remain algebraic.

Remark (Do not condense the mass first). It is tempting to eliminate the edge fields *already at the semidiscrete level* and to write a second-order nodal ODE $\bar{\mathcal{M}}(\partial_{tt}\boldsymbol{\lambda}, \boldsymbol{\mu}) + \bar{\mathcal{K}}(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \bar{F}(\boldsymbol{\mu})$ with the *static* condensed mass $\bar{\mathcal{M}}(\boldsymbol{\lambda}, \boldsymbol{\mu}) = c(\boldsymbol{Y}_\mathrm{st}(\boldsymbol{\lambda}), \boldsymbol{Y}_\mathrm{st}(\boldsymbol{\mu}))$ assembled from the static (elliptic) local solver $\boldsymbol{Y}_\mathrm{st}$. This is a *static condensation of the mass* (Guyan reduction). The semidiscrete primal field $\boldsymbol{y}$ satisfies a genuine second-order edge ODE $c\,\partial_{tt}\boldsymbol{y} + S\boldsymbol{y} = G(\boldsymbol{\lambda}, \boldsymbol{f})$: its interior degrees of freedom carry inertia and are *not* algebraically slaved to the endpoint traces, so $\boldsymbol{y} \neq \boldsymbol{Y}_\mathrm{st}(\boldsymbol{\lambda}) + \boldsymbol{Y}_\mathrm{st,D}$ and $\partial_{tt}\boldsymbol{Y}_\mathrm{st}(\boldsymbol{\lambda}) = \boldsymbol{Y}_\mathrm{st}(\partial_{tt}\boldsymbol{\lambda})$ holds only for the static map. Consequently $\bar{\mathcal{M}}$ reproduces the inertia of the trace-driven constraint modes but discards the edge-interior modes; the resulting mass error is $O(h^2)$ *independent of the polynomial degree* p, and it caps the spatial convergence at order two even for high p. We therefore keep the inertia at the field level $(\boldsymbol{y}, \boldsymbol{z})$ and condense *only after* the stage discretization below.

2 The fully implicit Gauß step

Collecting the dynamic fields $\boldsymbol{U} := (\boldsymbol{y}, \boldsymbol{z})$ and eliminating the algebraic $\boldsymbol{q}, \boldsymbol{\lambda}$ pointwise in time (a *static, elliptic* condensation, which touches only the stiffness and is exact) puts (3) in the linear first-order form

$$\partial_t \boldsymbol{U} = \mathbf{C}\,\boldsymbol{U} + \boldsymbol{g}. \tag{4}$$

Here $\mathbf{C}$ acts on the *field* space and, crucially, $\boldsymbol{U}$ retains *all* edge-interior inertial degrees of freedom — no mass has been condensed. We never form $\mathbf{C}$ explicitly; each stage below is assembled and condensed directly as a saddle system.

Let $(\mathbf{A}, b, c)$ be the s-stage Gauß tableau. The naive step solves the coupled stage system $\big[\mathbf{I} - \Delta t\,(\mathbf{A} \otimes \mathbf{C})\big]\boldsymbol{Z} = \dots$ with all s stages entangled over the whole field state. (For $s = 1$, $\mathbf{A} = \frac{1}{2}$, this is the implicit midpoint / Crank–Nicolson scheme of `main.tex`; the point here is the high-order case $s > 1$.)

3 Trick 1: one condensed bilinear form per stage

Diagonalize the Butcher matrix, $\mathbf{A} = \mathbf{T}\boldsymbol{\Theta}\mathbf{T}^{-1}$, $\boldsymbol{\Theta} = \mathrm{diag}(\theta_1, \dots, \theta_s) \in \mathbb{C}^{s\times s}$ (distinct nonzero eigenvalues, generally complex). Exactly as in `trick1.tex`, the change of variables $\boldsymbol{Z} = (\mathbf{T}^{-1} \otimes \mathbf{I})\boldsymbol{U}$ block-diagonalizes the stage system into s *independent* problems $(\mathbf{I} - \Delta t\,\theta_\ell \mathbf{C})\boldsymbol{Z}_\ell = \boldsymbol{r}_\ell$. Written out on the fields, stage ℓ is a complex saddle system *structurally identical to the fully discrete Crank–Nicolson system of `main.tex`*, with the step factor $\frac{\Delta t}{2}$ replaced by $\Delta t\,\theta_\ell$. With $\sigma_\ell := 1/(\Delta t\,\theta_\ell)$, it reads: find $(\boldsymbol{q}_\ell, \boldsymbol{y}_\ell, \boldsymbol{z}_\ell, \boldsymbol{\zeta}_\ell)$ with

$$
\begin{aligned}
a(\boldsymbol{q}_\ell, \boldsymbol{w}_\mathrm{q}) + b(\boldsymbol{w}_\mathrm{q}, \boldsymbol{y}_\ell) &= -\langle \boldsymbol{\zeta}_\ell, \boldsymbol{w}_\mathrm{q}\, \nu\rangle_\mathcal{E}, && \text{(5a)}\\
-\, b(\boldsymbol{q}_\ell, \boldsymbol{w}_\mathrm{y}) + \langle \tau(\boldsymbol{y}_\ell - \boldsymbol{\zeta}_\ell), \boldsymbol{w}_\mathrm{y}\rangle_\mathcal{E} + \sigma_\ell(\boldsymbol{z}_\ell, \boldsymbol{w}_\mathrm{y})_\Sigma &= \widehat{g}_{\mathrm{y},\ell}(\boldsymbol{w}_\mathrm{y}), && \text{(5b)}\\
-\, \sigma_\ell(\boldsymbol{y}_\ell, \boldsymbol{w}_\mathrm{z})_\Sigma + d(\boldsymbol{z}_\ell, \boldsymbol{w}_\mathrm{z})_\Sigma &= \widehat{g}_{\mathrm{z},\ell}(\boldsymbol{w}_\mathrm{z}), && \text{(5c)}
\end{aligned}
$$

together with the nodal balance condition.

The stage unknowns and the physical trace. The stage quantities are the *Butcher-eigenbasis coordinates* of the physical stage values. Writing $\boldsymbol{q}_j, \boldsymbol{y}_j, \boldsymbol{z}_j, \boldsymbol{\lambda}_j$ for the values at the collocation node $t^n + c_j\Delta t$ ($j = 1, \dots, s$, with c_j the Gauß nodes),

$$\boldsymbol{\zeta}_\ell = \sum_{j=1}^{s} (\mathbf{T}^{-1})_{\ell j}\, \boldsymbol{\lambda}_j, \qquad \boldsymbol{y}_\ell = \sum_{j=1}^{s} (\mathbf{T}^{-1})_{\ell j}\, \boldsymbol{y}_j, \qquad \boldsymbol{\lambda}_i = \sum_{\ell=1}^{s} \mathbf{T}_{i\ell}\, \boldsymbol{\zeta}_\ell, \tag{6}$$

and analogously for $\boldsymbol{q}_\ell, \boldsymbol{z}_\ell$. Thus $\boldsymbol{\zeta}_\ell$ is *not* a trace at a physical time but a complex combination of the s real stage traces $\boldsymbol{\lambda}_j$. Since $\mathbf{T}$ — the eigenvector matrix of the real, nonsymmetric $\mathbf{A}$ — is complex, $\boldsymbol{\zeta}_\ell \in \boldsymbol{\Lambda} \otimes \mathbb{C}$. This complexity is entirely an artifact of the diagonalization, not of the physics (the material data are real); Trick 2 recovers the real stage traces $\boldsymbol{\lambda}_i$ by conjugation.

The loads. Writing $\mathbb{1} = (1, \dots, 1)^\top$, $\omega_\ell := (\mathbf{T}^{-1}\mathbb{1})_\ell = \sum_j (\mathbf{T}^{-1})_{\ell j}$, and transforming the source by $\widetilde{\boldsymbol{f}}_\ell := \sum_j (\mathbf{T}^{-1})_{\ell j}\, \boldsymbol{f}(t^n + c_j\Delta t)$, the loads in (5) are explicitly

$$\widehat{g}_{\mathrm{y},\ell}(\boldsymbol{w}_\mathrm{y}) = (\widetilde{\boldsymbol{f}}_\ell, \boldsymbol{w}_\mathrm{y})_\Sigma + \sigma_\ell\, \omega_\ell\, (\boldsymbol{z}^n, \boldsymbol{w}_\mathrm{y})_\Sigma, \qquad \widehat{g}_{\mathrm{z},\ell}(\boldsymbol{w}_\mathrm{z}) = -\,\sigma_\ell\, \omega_\ell\, (\boldsymbol{y}^n, \boldsymbol{w}_\mathrm{z})_\Sigma, \tag{7}$$

where $(\boldsymbol{y}^n, \boldsymbol{z}^n)$ are the fields at the start of the step. *Only* the initial value $(\boldsymbol{y}^n, \boldsymbol{z}^n)$ enters as history: $\boldsymbol{q}$ and $\boldsymbol{\lambda}$ carry no time derivative and are recomputed algebraically at each stage (inhomogeneous Dirichlet data would enter these same loads through the splitting $\boldsymbol{\lambda} = \lambda_\mathrm{D} + \boldsymbol{\lambda}$). This is the collocation form of `trick1.tex`, eq. (3.7); for $s = 1$ ($\theta_1 = \frac{1}{2}$, $\omega_1 = 1$, $\sigma_1 = 2/\Delta t$) it is the diagonalized Crank–Nicolson step of `main.tex`.

The decisive difference from (3) is that in (5) the coupling $\partial_t \to \sigma_\ell$ is *algebraic*: restricting (5) to a single edge now defines genuine *stage-ℓ local solvers*, which invert the local compliance–equilibrium–stabilization *together with the σ_ℓ-weighted mass*:

$$\boldsymbol{Q}_\ell : \boldsymbol{\zeta}_\ell \mapsto \boldsymbol{q}_\ell, \qquad \boldsymbol{Y}_\ell : \boldsymbol{\zeta}_\ell \mapsto \boldsymbol{y}_\ell, \qquad \boldsymbol{Z}_\ell : \boldsymbol{\zeta}_\ell \mapsto \boldsymbol{z}_\ell, \tag{8}$$

together with their data parts $\boldsymbol{Q}_{\mathrm{D},\ell}, \boldsymbol{Y}_{\mathrm{D},\ell}, \boldsymbol{Z}_{\mathrm{D},\ell}$. Unlike the static solver of Remark 1, these *depend on θ_ℓ*: the reconstructed edge field $\boldsymbol{Y}_\ell(\boldsymbol{\zeta}_\ell)$ already carries the edge-interior inertia.

Proposition 1 (Per-stage condensed form)**.** *For each $\ell = 1, \dots, s$ the stage solve reduces to a single condensed problem on the nodal space: find $\boldsymbol{\zeta}_\ell \in \boldsymbol{\Lambda} \otimes \mathbb{C}$ with*

$$\boxed{\widehat{A}_\ell(\boldsymbol{\zeta}_\ell, \boldsymbol{\mu}) = \widehat{F}_\ell(\boldsymbol{\mu}) \qquad \forall \boldsymbol{\mu} \in \boldsymbol{\Lambda},} \tag{9}$$

where, in complete analogy to the fully discrete condensed operator $\widehat{A}$ of `main.tex` (with θ_ℓ in place of $\frac{1}{2}$),

$$\widehat{A}_\ell(\boldsymbol{\zeta}_\ell, \boldsymbol{\mu}) = a\big(\boldsymbol{Q}_\ell(\boldsymbol{\zeta}_\ell), \boldsymbol{Q}_\ell(\boldsymbol{\mu})\big) + d\big(\boldsymbol{Z}_\ell(\boldsymbol{\zeta}_\ell), \boldsymbol{Z}_\ell(\boldsymbol{\mu})\big) + \big\langle \tau(\boldsymbol{Y}_\ell(\boldsymbol{\zeta}_\ell) - \boldsymbol{\zeta}_\ell), \boldsymbol{Y}_\ell(\boldsymbol{\mu}) - \boldsymbol{\mu} \big\rangle_{\mathcal{E}}. \tag{10}$$

The form $\widehat{A}_\ell$ is complex-symmetric (not Hermitian) and invertible for $\tau > 0$. The right-hand side is

$$\widehat{F}_\ell(\boldsymbol{\mu}) = \langle \boldsymbol{Q}_{\mathrm{D},\ell}\, \nu + \tau\, \boldsymbol{Y}_{\mathrm{D},\ell}, \boldsymbol{\mu} \rangle_{\mathcal{E}}, \tag{11}$$

where the data fields $(\boldsymbol{Q}_{\mathrm{D},\ell}, \boldsymbol{Y}_{\mathrm{D},\ell}, \boldsymbol{Z}_{\mathrm{D},\ell})$ solve the stage system (5) with $\boldsymbol{\zeta}_\ell = 0$ and the loads (7).

Proof. Eliminating $\boldsymbol{q}_\ell, \boldsymbol{y}_\ell, \boldsymbol{z}_\ell$ from (5) through the local solvers (8) and inserting the result into the nodal flux-balance condition gives (9); testing the balance with the local reconstructions of $\boldsymbol{\mu}$ and adding the three edge equations yields the symmetric representation (10), exactly as in the fully discrete Lemma of `main.tex` with $\Delta t/2 \rightsquigarrow \Delta t\,\theta_\ell$. □

Remark (Contrast with the static-mass form). Eliminating $\boldsymbol{z}_\ell$ from (5) via its third equation, $\boldsymbol{z}_\ell = \sigma_\ell \mathbf{C}_{\mathrm{y}} \boldsymbol{y}_\ell + \dots$, turns (10) into the equivalent second-order-style operator

$$c\big(\boldsymbol{Y}_\ell(\boldsymbol{\zeta}_\ell), \boldsymbol{Y}_\ell(\boldsymbol{\mu})\big) + (\Delta t\,\theta_\ell)^2 \Big[a\big(\boldsymbol{Q}_\ell(\boldsymbol{\zeta}_\ell), \boldsymbol{Q}_\ell(\boldsymbol{\mu})\big) + \big\langle \tau(\boldsymbol{Y}_\ell(\boldsymbol{\zeta}_\ell) - \boldsymbol{\zeta}_\ell), \boldsymbol{Y}_\ell(\boldsymbol{\mu}) - \boldsymbol{\mu} \big\rangle_{\mathcal{E}} \Big]. \tag{12}$$

This has the *same shape* as the naive per-stage form $\bar{\mathcal{M}} + (\Delta t\,\theta_\ell)^2 \bar{\mathcal{K}}$ of the earlier draft, but with the *stage-dependent, mass-shifted* local solvers $\boldsymbol{Q}_\ell, \boldsymbol{Y}_\ell$ in place of the *static* $\boldsymbol{Q}, \boldsymbol{Y}$. Solving the local problem with the mass included — rather than reusing static local solvers and adding a static condensed mass — is precisely what restores the optimal spatial order h^{p+1}. The Butcher diagonalization therefore still *commutes with the condensation*, but the condensation must be carried out *per stage, after* θ_ℓ is fixed, so that the mass sits inside the local solve.

Remark (Schur reduction; how many fields can be eliminated). The second-order form (12) is a *Schur complement* of the stage saddle (5): the velocity field $\boldsymbol{z}_\ell$ is eliminated through the third equation ($d(\boldsymbol{z}_\ell, \cdot) = \sigma_\ell(\boldsymbol{y}_\ell, \cdot) + \widehat{g}_{\mathrm{z},\ell}$, i.e. $\boldsymbol{z}_\ell = \sigma_\ell \mathbf{C}_{\mathrm{y}} \boldsymbol{y}_\ell + \dots$) and recovered explicitly afterwards, leaving the complex weight $\alpha_\ell := (\Delta t\,\theta_\ell)^2 \in \mathbb{C}$ in front of the stiffness. This is exactly the last reduction of Hochbruck, Köhler & Kumbhar (*Preconditioned implicit time integration schemes for Maxwell's equations*, CRC 1173 preprint 2022/29, eqs. (3.12)–(3.14)): their first-order Maxwell pair $(\boldsymbol{H}, \boldsymbol{E})$ decouples per stage into a size-$2N$ system $(\mathbf{I} - \Delta t \theta_\ell \mathbf{C}) \boldsymbol{Z}_\ell = \boldsymbol{r}_\ell$, whose Schur complement in one field is $(\mathbf{I} + \alpha_\ell\, \mathbf{C}_{\mathrm{H}} \mathbf{C}_{\mathrm{E}}) \boldsymbol{Z}_{\mathrm{E},\ell} = \dots$ of size N, the partner $\boldsymbol{Z}_{\mathrm{H},\ell}$ following explicitly. Here $\mathbf{C}_{\mathrm{H}} \mathbf{C}_{\mathrm{E}}$ is the condensed stiffness $a(\boldsymbol{Q}_{\ell'} \cdot, \boldsymbol{Q}_{\ell'} \cdot) + \langle \tau(\cdot), \cdot \rangle$ and the mass $c(\boldsymbol{Y}_{\ell'} \cdot, \boldsymbol{Y}_{\ell'} \cdot)$ plays the role of their identity.

Counting eliminations: of the three edge fields, *two* leave the global solve. The dual $\boldsymbol{q}_\ell$ is algebraic and is removed by the local elliptic solver $\boldsymbol{Q}_\ell$; one of the two dynamical fields ($\boldsymbol{z}_\ell$) is removed by the Schur step and recovered explicitly; the remaining field $\boldsymbol{y}_\ell$ is condensed to the skeleton. What survives is a *single* complex-symmetric nodal system (9) per representative stage. One cannot remove *both* dynamical fields — each edge keeps one genuine (second-order) degree of freedom — so three-to-one is the maximal reduction. As in Remark 3, the Schur step must be taken on the mass-shifted saddle (5) (its solvers carry θ_ℓ); taking it on the static solvers would reinstate the order reduction of Remark 1.

4 Trick 2: conjugate pairs

Since $\mathbf{A}$, the material forms, and the data are real, the eigenvalues θ_ℓ, hence the shifts $\sigma_\ell = 1/(\Delta t\,\theta_\ell)$, occur in complex-conjugate pairs. The local solvers and the condensed operator inherit this symmetry, $\boldsymbol{Q}_{\ell'} = \overline{\boldsymbol{Q}_\ell}$, $\boldsymbol{Y}_{\ell'} = \overline{\boldsymbol{Y}_\ell}$, $\boldsymbol{Z}_{\ell'} = \overline{\boldsymbol{Z}_\ell}$ and $\widehat{A}_{\ell'} = \overline{\widehat{A}_\ell}$ whenever $\theta_{\ell'} = \overline{\theta_\ell}$, so the stage solutions satisfy $\boldsymbol{\zeta}_{\ell'} = \overline{\boldsymbol{\zeta}_\ell}$ (Lemma 3.1 of `trick1.tex`). One therefore assembles and factorizes the condensed nodal operator $\widehat{A}_\ell$ for only $\lceil s/2 \rceil$ representatives and obtains the conjugate partners for free.

5 Algorithm, recovery, and cost

One fully implicit, order-$2s$ Gauß step on (3) is:

1. **Transform** the stage data by $\mathbf{T}^{-1}$ to form the loads $\widehat{g}_{\mathrm{y},\ell}, \widehat{g}_{\mathrm{z},\ell}$ and hence $\widehat{F}_\ell$.

2. **Nodal solves:** for each representative ℓ ($\lceil s/2 \rceil$ of them) assemble $\widehat{A}_\ell$ from the θ_ℓ-shifted local solvers $\boldsymbol{Q}_\ell, \boldsymbol{Y}_\ell, \boldsymbol{Z}_\ell$ and solve the condensed problem (9) on $\boldsymbol{\Lambda}$; conjugate the partners.

3. **Back-transform** $\boldsymbol{U}_i = \sum_\ell \mathbf{T}_{i\ell}\,(\,\cdot\,)_\ell$ to recover the stage values $(\boldsymbol{y}_i, \boldsymbol{z}_i)$ (real up to round-off) and apply the Gauß update for $(\boldsymbol{y}^{n+1}, \boldsymbol{z}^{n+1})$ and the trace.

4. **Recover the edge fields** where needed via the local solvers, $\boldsymbol{q} = \boldsymbol{Q}_\ell(\boldsymbol{\zeta}_\ell) + \boldsymbol{Q}_{\mathrm{D},\ell}$ etc. — the edge interior never enters the global solve.

Remark (Cost and reuse). For fixed Δt the shifts σ_ℓ are fixed complex numbers, so the $\lceil s/2 \rceil$ condensed operators $\widehat{A}_\ell$ are assembled from the θ_ℓ-shifted local solvers and factorized *once*, and reused at every step. Complex arithmetic is confined to these small nodal systems and the edgewise local solves. Unlike the earlier draft, one does *not* pre-assemble a single real pair $\bar{\mathcal{M}}, \bar{\mathcal{K}}$ and combine them as $\bar{\mathcal{M}} + (\Delta t\,\theta_\ell)^2 \bar{\mathcal{K}}$: the local solvers depend on θ_ℓ because they invert stiffness *and* mass. The per-step cost is nevertheless the same — $\lceil s/2 \rceil$ condensed solves, each factorized once — and the naive step would instead couple all s stages into one real system over the whole field state.

Remark (Order of condensation and time discretization). Two orderings are possible. **(a) Condense first:** reduce the semidiscrete problem to the nodal second-order ODE $\bar{\mathcal{M}}(\partial_{tt}\boldsymbol{\lambda}, \boldsymbol{\mu}) + \bar{\mathcal{K}}(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \bar{F}(\boldsymbol{\mu})$ with a static condensed mass (Remark 1), then apply Gauß. **(b) Condense last** (this note, and `main.tex`): discretize the first-order field system (3) in time/stages, then condense each stage, (9). For $p = 0$, where the edges carry no interior modes, the two orderings coincide. For $p \geq 1$ they *differ* by the edge-interior inertia: ordering (a) statically condenses the mass, and its spatial error saturates at $O(h^2)$ regardless of p; ordering (b) keeps the interior inertia inside the stage-ℓ local solve and attains the optimal rate h^{p+1} established for the Crank–Nicolson scheme ($s = 1$) in `main.tex`. An earlier version of this note used ordering (a); the present version uses (b). This is the "one global condensed bilinear form per stage" anticipated in the Remark "Choice of time discretization" of `main.tex`, now with the inertia correctly retained.